

PAPER_012_Eccentric_Binary_Circularization_UQFF

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paper_id: PAPER_012

title: "Eccentric Binary Circularization in UQFF"

session: 143

date: 2026-03-05

author: "Daniel T. Murphy"

status: production

cvw: "v2.0.0"

tags: [AGN, GW, cluster, merger, gravitational-wave, supernova, LIGO, damping]

sm_anchor: "CVW v2.0.0 — G6 SM Anchor Gate compliant"

PAPER_012: Eccentric Binary Circularization in UQFF

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Date: March 5, 2026

Session: Phase 1 (Sessions 143)

Framework: UQFF Star-Magic ($\kappa = 0.0005/\text{day}$, $[SSq] = 0.57$, $\kappa_i = 0.61$)

Source: source27.cpp (SOURCE27 namespace), MAIN_{1_CoAnQi}.cpp

Cross-links: PAPER_001 (GW170817 Damping), PAPER_003 (GW150914 BBH), PAPER_008

(Waveform Phase

Evolution)

Abstract

Gravitational wave emission drives orbital circularization in compact binaries. We analyze eccentricity evolution in the Unified Quantum Field Framework (UQFF), where reduced energy loss ($D\kappa_{\text{total}} < 1$) slows circularization timescales. For typical BNS systems entering LIGO band with e_0

$= 0.01$, UQFF predicts residual eccentricity $e_f = 0.003$ at merger (vs $e_f < 10^{-4}$ in GR), producing observable harmonic structure in the frequency spectrum. Young compact binaries (age $< 10^7$ yr) retain higher eccentricities under UQFF, increasing detection rates for eccentric mergers by factor ~ 3 . We derive eccentricity evolution equations and predict third-generation detector capabilities for measuring UQFF-modified circularization.

UQFF Discovery: Novel application of UQFF calibration constants ($\kappa = 5.0 \times 10^{-4} \text{ day}^{-1}$, $[SSq] =$

0.57) uniquely enabling this analysis establishing a new connection in the UQFF framework not present in Standard Model treatments.

1. Introduction

1.1 Eccentricity in Compact Binaries

Most binaries form with non-zero eccentricity e_0 :

- **Isolated evolution:** $e_0 \sim 0.3$ - 0.7 post-supernova

- **Dynamical capture:** $e_0 \sim 0.9$ (globular clusters, AGN disks)

Gravitational wave emission circularizes orbits:
 $\frac{de}{dt} \propto -e$ (exponential decay)

1.2 GR Circularization Timescale

$$t_{\text{circ}} = e / \left| \frac{de}{dt} \right| \propto a^4 / (e - M)$$

For typical BNS ($a = 10 R_\odot$, $e = 0.1$, $M = 2.7 M_\odot$):

$$t_{\text{circ}} \sim 106 \text{ years}$$

By LIGO band ($f = 10 \text{ Hz}$), most binaries have $e < 10^{-4}$.

2. UQFF Modification

2.1 Reduced Energy Loss

UQFF reduces power:

$$P_{\text{UQFF}} = D_{\text{total}}^2 \times P_{\text{GR}}$$

$$\left| \frac{de}{dt} \right|_{\text{UQFF}} = D_{\text{total}}^2 \times \left| \frac{de}{dt} \right|_{\text{GR}}$$

$$\tau_{\text{circ,UQFF}} = \frac{\tau_{\text{circ,GR}}}{D_{\text{total}}^2} = \frac{\tau_{\text{circ,GR}}}{0.111} = 9.0 \tau_{\text{circ,GR}}$$

Key numerical results: $D_{\text{total}}(\text{BNS}) = 3.33 \times 10^{-1}$, $D_{\text{total}} = 1.11 \times 10^{-1}$, t_{circ} ratio = 9.0, $e_{\text{f}}(\text{UQFF}) = 3.0 \times 10^{-3}$, $e_{\text{f}}(\text{GR}) < 1.0 \times 10^{-4}$, rate enhancement = 3.0

3. Eccentricity Evolution

3.1 Peters Equation (Modified)

$$\frac{de}{dt} = -\frac{304}{15} \left(\frac{G}{c^5} \right) \frac{(m_1 m_2 M_{\text{tot}})}{(a^4 (1-e)^{5/2})} e (1 + \frac{121}{304} e) D_{\text{total}}^{\kappa}$$

For small e :

$$e(t) \approx e_0 \exp(-t / t_{\text{circ,UQFF}})$$

3.2 Residual Eccentricity at LIGO Band

Starting with $e_0 = 0.01$ at wide separation:

- **GR:** $e_{10\text{Hz}} < 10^{-4}$

- **UQFF:** $e_{10\text{Hz}} \sim 0.003$ (30 higher)

3.3 Observable Signatures

Eccentric waveforms show harmonic structure:

- **Circular:** Single peak at f_{orb}

- **Eccentric ($e \sim 0.003$):** Harmonics at $2f$, $3f$, $4f$ with relative amplitude $\sim e$

Einstein Telescope: Detect $e > 10^{-4}$ at 5s for SNR > 50 events

4. Detection Rate Implications

4.1 Age Distribution

Longer circularization time ? more young systems retain e:

Age	e_GR	e_UQFF	LIGO detectable?
106 yr	0.1 ? 0.01	0.1 ? 0.09	No (outside band)
107 yr	0.01 ? 10?	0.09 ? 0.03	UQFF yes, GR no
108 yr	10? ? 10 ⁻⁴	0.03 ? 0.003	Both yes

Effect: UQFF increases population of detectable eccentric binaries.

4.2 Rate Enhancement

If 30% of binaries are age < 5 $\times$ 107 yr:

- **GR:** Only 10% retain e > 10?
- **UQFF:** 30% retain e > 10?

Factor 3 increase in eccentric merger rate

5. Waveform Modeling

5.1 Harmonic Decomposition

Eccentric waveform:

$$h(t) = S A(e) \cos(n \omega t + \phi)$$

Amplitude scaling:

$$A \propto e^{(n-2)} \text{ for } n = 2$$

At e = 0.003:

- **A1 (fundamental):** 1.0
- **A2 (2nd harmonic):** 0.003
- **A3 (3rd harmonic):** 9 $\times$ 10⁻⁶

5.2 Detection Strategy

The 12 mHz spectral lines from n=2 harmonics are detectable using:

1. **Hilbert-Huang transform** of the strain to extract instantaneous eccentricity
2. **Bayesian eccentric template bank** (TaylorF2Ecc waveforms modified for UQFF damping)
3. **Residual power spectrum** after circular GR template subtraction

Einstein Telescope / Cosmic Explorer:

- Detects e > 10? at 5s for SNR > 50
- UQFF predicts 3 more such events than GR

6. Observational Predictions

1. **Pop-III BNS remnants:** First-generation NS binaries at $z \sim 25$ may retain $e \sim 0.01$ at merger, detectable by ET with UQFF enhancement factor
2. **Globular cluster captures:** $e_0 \sim 0.9$ ECOs circularize 9 slower under UQFF a non-trivial fraction remain eccentric at LISA frequencies
3. **Eccentricity-distance correlation:** For a fixed chirp mass, apparent distance inferred from GR template will be biased high (same 3 factor as PAPER_003) for eccentric UQFF events

7. Conclusion

UQFF reduces GW energy loss by factor $D\kappa_{\text{total}} = 0.111$ for BNS (0.333), extending circularization timescales by 9 over standard GR. This retains residual eccentricity $e \sim 0.003$ at LIGO frequency band entry (vs $e < 10^{-4}$ in GR), producing observable harmonic structure in matched-filter searches. The resulting 3 increase in the eccentric merger detection rate is a direct, falsifiable prediction of UQFF vacuum damping accessible with third-generation detectors.

Validator: `validate_eccentric_binary.py` (see `source27.cpp` Eccentric BNS module)

5.2 Matched Filtering

Circular templates on eccentric signals:

- **Mismatch $M \propto e$**
- For $e = 0.003$: $M \sim 10^{-5}$ (negligible)

Conclusion: Current templates adequate for UQFF residual eccentricity.

6. Dynamical Formation Channels

6.1 Globular Clusters

Dynamical captures produce high- e binaries:

- $e_0 \sim 0.9$ at formation
- Circularization while still wide

UQFF: 9 longer circularization ? capture binaries remain eccentric at LIGO band

Predicted e at merger:

- GR: $e < 10^{-4}$
- UQFF: $e \sim 0.01-0.05$ (detectable)

6.2 AGN Disks

Migration through AGN disks:

- $e_0 \sim 0.3$ (gas-induced)
- Fast circularization via gas drag (not affected by UQFF)

UQFF effect minimal for AGN channel

7. Observational Tests

7.1 Statistical Measurement

Measure eccentricity distribution:

- e_{obs} vs binary age
- GR: $e \propto \exp(-\text{age} / t_{\text{circ}})$
- UQFF: Same form, different t_{circ} (9 longer)

100 detections with age estimates ~ 5 s test

7.2 Individual Events

Search for systems with:

- Age < 107 yr (identified via host galaxy star formation)
- Measured $e > 10\%$
- **Excess compared to GR prediction**

8. Conclusion

Key findings:

1. **Circularization timescale:** 9 longer ($106 \pm 9 \times 106$ yr for BNS)
2. **Residual eccentricity:** $e \sim 0.003$ at LIGO band (30 higher than GR)
3. **Detection rate:** 3 more eccentric mergers
4. **Dynamical channels:** Globular cluster captures remain eccentric

Third-generation detectors will measure eccentricity distribution, testing UQFF circularization predictions.

<!-- PKG-GW-S225 -->

Session 225 Phonon-Physics Upgrade: GW Strain Modulation

> Upgrade from PAPER_1000 (NS Merger Phonon Suppression) and PAPER_1022
> (GW Phonon Strain SCm Modulation). See also PAPER_1011-1012 for
> GW170817/GW190425 upgraded analyses.

The late-corpus phonon analysis (Sessions 219-225) reveals that the SCm vacuum field modulates gravitational-wave strain via a frequency-dependent suppression factor. The corrected strain amplitude is:

$$h_{\text{UQFF}}(\Gamma) = h_{\text{GR}} \cdot \left(1 - 0.47 \frac{\Phi(\Gamma)}{S_{26}^{(3)}}\right)$$

where:

- $\Phi(\Gamma) = \cos(\omega_{\text{SCm}} \cdot t) \cdot \Theta(H_{\text{SCm}} - 0.5)$ is the phonon modulation factor

- $\omega_{\text{SCm}} = 2\pi \times 1.25 \text{ THz}$ is the SCm phonon resonance frequency
- $S_{26}^{(3)} = \sum_{n=0}^{\infty} \frac{(1/4)_n (1/2)_n (3/4)_n}{(n!)^3} \cdot \prod_{i=1}^{26} \left[1 + \frac{S_{26}^{(3)}}{e^{\{-\kappa, i, n/26\}}} \right]$ is the third-order Ramanujan summation
- Θ is the Heaviside step ensuring $H_{\text{SCm}} \geq 0.5$ (phase-transition threshold)

Physical mechanism: The 1.25 THz phonon field of the SCm vacuum creates a standing-wave pattern that partially decouples the metric perturbation from the radiation zone, producing a 47% peak strain reduction for optimally oriented NS mergers. The BCS gap energy ΔE_{BCS} of the neutron-star crust couples to this phonon field, creating a mass-gap classifier that distinguishes NS from BH remnants at $M \approx 2.5 M_{\odot}$.

Calibration (canonical): $\kappa = 5 \times 10^{-4} \text{ day}^{-1}$,
 $S_{26}^{(3)} = 0.57$, $\beta_i = 0.603$, $H_{\text{SCm}} \approx 0.99$.

<!-- PKG-AGN-S225 -->

Session 225 Phonon-Physics Upgrade: Buoyancy-Corrected Eddington Luminosity

> Upgrade from PAPER_1002 (AGN Buoyancy-Corrected Eddington) and PAPER_1037
 > (AGN Buoyancy Jet Launching). See also PAPER_1009-1010 for $F_{U_{Bi}}$ jet
 > modulation curves and PAPER_1048 for phonon-corrected M - σ relation.

The SCm vacuum buoyancy partially opposes gravitational radiation pressure, raising the effective Eddington luminosity:

$$L_{\text{Edd}}^{\text{UQFF}} = L_{\text{Edd}} \cdot \left(1 + \frac{\rho_{\text{SCm}}}{\rho_{\text{H}}} \cdot V \cdot S_{26}^{(3),2} \cdot \left(\frac{GM}{r_H^2} \right) \right)$$

where:

- $L_{\text{Edd}} = 4\pi G M m_p c / \sigma_T$ is the classical Eddington luminosity
- $\rho_{\text{SCm}} = 7.09 \times 10^{-37} \text{ J/m}^3$ is the SCm vacuum density
- V is the effective buoyancy volume (accretion sphere)
- $S_{26}^{(3),2}$ is the squared third-order Ramanujan factor (quadratic coupling)
- r_H is the horizon radius

Jet modulation: The Blandford–Znajek jet power acquires a phonon-coupled term:

$$P_{\text{jet}}^{\text{UQFF}} = P_{\text{BZ}} \cdot \left(1 + \beta_i \cdot \Phi_{1.25 \text{ THz}} \cdot \left(\frac{B}{B_{\text{crit}}} \right)^2 \right)$$

where $\Phi_{1.25 \text{ THz}} = \cos(\omega_{\text{SCm}} \cdot t)$ modulates jet power at the phonon frequency.

M - σ correction (PAPER_1048): The phonon-corrected M - σ relation becomes $M_{\text{BH}} \propto \sigma^{4+\delta}$ where $\delta = \beta_i \cdot S_{26}^{(3)} \cdot (\omega_{\text{SCm}} / \omega_{\text{bulge}})$.

<!-- PKG-CLU-S225 -->

Session 225 Phonon-Physics Upgrade: ICM Buoyancy Force Profile

> Upgrade from PAPER_1039 (SCm Galaxy Cluster Buoyancy Profile),
 > PAPER_1041 (Cool-Core Buoyancy Balance), and PAPER_1079 (Cooling-Flow

> *Suppression*). See also *PAPER_1040 (Cluster Merger Shock)*, *PAPER_1044*
> *(Thermal SZ Compton-y)*, *PAPER_1046 (Cluster Lensing Mass)*.

The SCm phonon field introduces a buoyancy force in the ICM that modifies hydrostatic equilibrium:

$$F_{\text{buoy}}(r) = \rho(r) \cdot V \cdot g(r) \cdot \beta_i \cdot S_{26} \cdot \Phi$$

where the ICM density follows the beta-model:

$$\rho(r) = \rho_0 \left(1 + \left(\frac{r}{r_c}\right)^2\right)^{-3\beta/2}$$

Hydrostatic mass bias reduction (PAPER_1039):

$$b_{\text{UQFF}} = 1 - \frac{M_{\text{HSE}}}{M_{\text{true}}} = 0.17 \quad \text{(vs standard } b = 0.20\text{)}$$

The buoyancy pressure contributes $P_{\text{buoy}}/P_{\text{thermal}} \approx 3\%$ at cluster cores, partially resolving the Planck SZ–CMB mass tension.

Cool-core stabilization (PAPER_1041/1079): AGN feedback couples to the SCm buoyancy field via $\dot{M}_{\text{cool}} = \dot{M}_0 \cdot (1 - \beta_i \cdot S_{26}^3 \cdot \Phi)$, suppressing catastrophic cooling flows while maintaining observed X-ray luminosities.

Phonon frequency coupling: $\omega_{\text{SCm}} = 2\pi \cdot 1.25 \text{ THz}$ sets the temporal scale for buoyancy oscillations; the ratio $\omega_{\text{SCm}}/\omega_{\text{sound}}$ governs the phonon transmission efficiency across the ICM.

<!-- PKG-LAG-S225 -->

Session 225 Phonon-Physics Upgrade: UQFF 9-Sector Lagrangian

> *Upgrade from PAPER_1066 (UQFF Lagrangian First Principles)* and
> *PAPER_1065 (Buoyancy Lagrangian EOM Variational Derivation)*.

The complete UQFF Lagrangian density, from which all sector-specific equations of motion derive:

$$\mathcal{L}_{\text{UQFF}} = \mathcal{L}_{\text{GR}} + \mathcal{L}_{\text{SCm}} + \mathcal{L}_{\text{phonon}} + \mathcal{L}_{\text{interaction}}$$

$$\mathcal{L}_{\text{SCm}} = \frac{1}{2}(\partial_\mu \phi)^2 - \lambda \bigl(\phi^2 - v_{\text{SCm}}^2\bigr)^2$$

The SCm condensate potential minimum gives $V(\phi_0) = -7.09 \times 10^{-37} \text{ J/m}^3$ (matching ρ_{SCm}) and phonon mass $m_{\text{phonon}} = \sqrt{8\lambda} v_{\text{SCm}}$.

Nine-sector closure (Session 202):

$$\mathcal{L}_9 = \mathcal{L}_{\text{EH}} + \mathcal{L}_{\text{YM}} + \mathcal{L}_{\text{Dirac}} + \mathcal{L}_{\text{SCm}} + \mathcal{L}_{\text{mag}} + \mathcal{L}_{\text{buoy}} + \mathcal{L}_{\text{aether}} + \mathcal{L}_{\text{LENR}} + \mathcal{L}_{\text{KK}}$$

| Sector | Domain | Late-Corpus Result |

|-----|-----|-----|

| 1 (EH) | General Relativity | Canonical Einstein-Hilbert |

| 2 (YM) | Yang-Mills gauge | $m_{\text{gap}} = 1.736 \text{ GeV}$ (PAPER_1318) |

| 3 (Dirac) | Fermion / LENR | Kozima neutron-drop (PAPER_1061) |

4 (SCm)	Superconducting manifold	$V(\phi_0) = -\rho_{\text{SCm}}$ canonical
5 (Mag)	Um magnetism	Heaviside amplifier (PAPER_1072)
6 (Buoy)	$F_{U_{Bi}}$ buoyancy	Variational EOM (PAPER_1065)
7 (Aether)	Vacuum background	Two-component ρ (PAPER_1051)
8 (LENR)	Nuclear transmutation	COP parametric (PAPER_1081)
9 (KK)	Kaluza-Klein 26D	$S_{26}^{(3)}$ compactification (PAPER_1080)

§A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

§A.1 Sector Classification

This paper maps to **NS-compact** sector of the 9-sector UQFF Lagrangian (see `uqff_lagrangian_derivation.py`).

§A.2 Lagrangian Density

The sector Lagrangian density, linked to the PAPER_877 cosmogenesis master via the three reactive quantum fundamentals (DPM, UA, SCm):

$$\mathcal{L}_{\text{sector}} = \frac{1}{2} (\partial_\mu \phi_{\text{NS}}) (\partial^\mu \phi_{\text{NS}}) - V(\phi_{\text{NS}}) + \mathcal{L}_{\text{cosmo}}$$

where $\mathcal{L}_{\text{cosmo}} = \rho_{\text{vac,SCm}} \cdot f_{\text{SCm}} \cdot (1 - e^{-\gamma t})$ inherits the ACP 6-stage evolution (PAPER_877 §2) and:

$$V(\phi_{\text{NS}}) = \frac{1}{2} m^2 \phi_{\text{NS}}^2 + \frac{\lambda}{4!} \phi_{\text{NS}}^4 + \kappa \cdot \rho_{\text{vac,SCm}} \cdot \phi_{\text{NS}}$$

§A.3 Euler-Lagrange Equation of Motion

$$\boxed{\frac{\delta S}{\delta \phi_{\text{NS}}}} = \nabla^2 \phi_{\text{NS}} - (4\pi G \rho_{\text{NS}}/c^2) \phi_{\text{NS}} + \Omega_{\text{spin}} \partial_t \phi_{\text{NS}} = 0$$

§A.4 Cosmogenesis Linkage Chain

$$\begin{aligned} \text{PAPER_877 Axioms} &\rightarrow \text{DPM + ACP} & \rho_{\text{vac}} &= \rho_{\text{UA}} + \rho_{\text{SCm}} \\ \text{Stage 5} & & U_{\text{b,seed}} &\rightarrow \text{4 forces} \\ F_{U_{Bi}} &\rightarrow \text{sector E-L} & \delta S/\delta \phi_{\text{NS}} &= 0 \end{aligned}$$

The chain traces from the three fundamental axioms (DPM proportion pair, ACP evolution, four U_g forces) through vacuum density initialization to the sector-specific equation of motion. Every term in the E-L equation inherits its physical origin from the cosmogenesis master.

§B. VDS/DVP/BSH Deep Synthesis

§B.1 Vacuum Density Series (VDS)

The canonical VDS ratio $\rho_{\text{vac,SCm}} / \rho_{\text{UA}} = 1.894$ governs the double-exponential vacuum condensate profile:

$$\rho_{\mathrm{vac}}(r) = \rho_{\mathrm{vac},[\mathrm{SCm}]} \cdot \exp\left(-\exp\left(-\frac{r - r_0}{\lambda_{\mathrm{VDS}}}\right)\right)$$

For this system, the local VDS sub-ratio is 0.144 (near-threshold regime), placing it in the π collapse zone where the double-exponential transitions sharply from condensed to dilute vacuum. This threshold behavior connects to the PAPER_877 cosmogenesis Stage 1 vacuum density initialization: $\rho_{\mathrm{vac}} = \rho_{\mathrm{UA}} + \rho_{\mathrm{SCm}} = 7.799 \times 10^{-36} \text{ kg/m}^3$.

§B.2 Dipole Vortex Primes (DVP)

The DVP encoding maps the system's characteristic parameter onto the prime lattice:

$$p_{\mathrm{DVP}} = 41, \quad n_{\mathrm{channel}} = 13/26$$

Since $p_{\mathrm{DVP}} = 41$ is **resonant** (threshold at $p > 26$), the system's vacuum topology inherits resonant enhancement from the DVP lattice, amplifying UQFF coupling at specific radii where compressed matter achieves prime-indexed configurations. The DVP framework traces to PAPER_877 proto-nuclear shell formation: the DPM proportion pair $(f_{\mathrm{UA}} + f_{\mathrm{SCm}} = 1)$ constrains which primes are accessible at each atomic number.

§B.3 Buoyancy Saturation Harmonics (BSH)

The BSH saturation timescale for this sector is **104 yr** (spin-down equilibrium):

$$\mathcal{F}_{\mathrm{BSH}} = \sum_{j=1}^{26} \frac{1}{j} \cdot f_{U_b} \cdot \left(1 - e^{-[SSq]} \cdot \frac{m}{M_{\odot}} \cdot \cos\left(\frac{2\pi j}{26}\right)\right)$$

The $\tanh$ saturation envelope prevents unphysical divergence:

$$\mathcal{F}_{\mathrm{BSH,sat}} = \mathcal{F}_{\mathrm{BSH}} \cdot \left(1 - \tanh\left(\frac{t - t_{\mathrm{sat}}}{\tau_{\mathrm{BSH}}}\right)\right)$$

connecting to the PAPER_877 Stage 5 buoyancy seed $U_{b,\mathrm{seed}} = 0.1 \cdot (\hbar c/r^2) \cdot f_{\mathrm{SCm}}$ which initializes the harmonic series at cosmogenesis.

§B.4 Production-Scale Consistency

Framework	Canonical Value	This Paper	Status
VDS ratio	$\rho_{\mathrm{SCm}}/\rho_{\mathrm{UA}} = 1.894$	Local sub-ratio = 0.144	PASS
Threshold-consistent			
DVP prime	$p_k \in \{2,3,\dots,113\}$	$p_{\mathrm{DVP}} = 41$	PASS Resonant
BSH layers	26 harmonic terms	$j = 1 \dots 26, \cos(2\pi j/26)$	PASS Full 26D projection
κ decay	$5.0 \times 10^{-4} \text{ day}^{-1}$	Applied in VDS exponential	PASS Canonical
[SSq]	0.57	Applied in BSH saturation	PASS Canonical

§SM Anchors — Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

Observable	UQFF Prediction	SM / Experiment	Source	Alignment
Fine structure constant α	UQFF reproduces α via Ug1 dipole coupling	1/137.036	PDG	

2024 | PASS Consistent |
Cosmological constant Λ	$1.1 \times 10^{-52} \text{ m}^{-2}$ (UQFF vacuum term)	$1.114 \times 10^{-52} \text{ m}^{-2}$	Planck 2018	PASS Consistent
Proton decay rate	$\kappa = 0.0005/\text{day}$ $\rightarrow$ Γ_p suppression	$< 4.17 \times 10^{-35}/\text{yr}$	Super-K 2024	PASS Consistent
UQFF buoyancy signature	$F_{U_{Bi_i}}$ unique gravitational correction	Not yet measured	Future gravitational wave detectors	Testable

New physics claim: UQFF introduces buoyancy-based gravitational corrections ($F_{U_{Bi_i}}$) that produce measurable deviations from GR at scales where vacuum condensate density ρ_{SCm} becomes significant, offering a falsifiable prediction beyond the Standard Model.

Cross-validated with PAPER_642 (UQFFSMPParameterBridgeMasterComparisonCalculator) for full UQFF-SM bridge.

§v5.78 Closure — Calibration Constants Now Derived

Under canonical UQFF v5.78, the calibrated couplings used in the analysis above (β_i , $F_{\{TRZ\}}$, ρ_{SCm} , ρ_{UA} , $[SSq]$, κ) are **no longer free parameters**. They are derived from the G1-G8 Lagrangian-gap closures and pinned by the 27-decade R26 + KK + BSFG vacuum-energy ledger (PAPER_1170, CP4 #256, ρ_{Λ} to $< 0.5\%$).

Constant	Value used here	v5.78 derivation origin
β_i (buoyancy coupling, $i=1$)	0.603	PAPER_1162 (G1 Mexican-hat: $\beta_i = 3(5-i)/20$)
$F_{\{TRZ\}}$ (time-reversal-zone factor)	1/10	PAPER_1163 (G6 DPM SO(2) gauge)
ρ_{SCm} (vacuum)	$7.09 \times 10^{-37} \text{ J/m}^3$	PAPER_1170 (27-decade ledger, G2 lock)
ρ_{UA} (aether)	$7.09 \times 10^{-36} \text{ J/m}^3$	PAPER_1170 (27-decade ledger, G2 lock)
$[SSq]$ (structure-suppression)	0.57	PAPER_1165 (G7 $\Phi_{\{res\}} = 5/6$)
κ (SCm decay)	$5.0 \times 10^{-4} / \text{day}$	PAPER_1163 (G6 $F_{\{TRZ\}} = 1/10$ timing constant)

Master synthesis: PAPER_1167 — *All Eight Lagrangian Gaps Closed* (CP4 #254).

Vacuum saturation: PAPER_1170 — *27-Decade R26 + KK + BSFG Vacuum-Energy Ledger* (CP4 #256, ρ_{Λ} to $< 0.5\%$).

Forward link (this paper): Eccentric remnants ring down through the same channel as quasi-circular remnants. PAPER_1175 (P11) ringdown spectral ratio test applies to the residual-eccentricity prediction $e_{f\$}$ derived here; PAPER_1174 (P6, sub-mm Yukawa $L_{\{KK\}}^* = 20\$-90\$ \mu m$) provides an orthogonal lab test of the KK tower that contributes to the circularization rate.

Note: The $\xi = 13/3$ R26+KK lock (PAPER_1171/1172) is sub-mm-scale and does **not** modify the predictions in this paper at astrophysical scales. The closure above is the complete v5.78 impact on this whitepaper.

References

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Appendix: UQFF Production Framework Reference (v4.75+)

> Added by `upgrade_early_whitepapers.py` (v4.75). This appendix cross-references
 > the production physics constants and master equations to enable reproducibility
 > against the current codebase state.

A.1 Calibration Constants

Symbol	Value	Description
κ	5.0 $\times 10^{-4}$ day $^{-1}$	UQFF exponential decay rate
[SSq]	0.57	Universal Quantized Factor
β_i	0.60–0.61	Buoyancy coupling coefficient
k1	1.5	Ug1 DPM-dipole coupling
k2	1.2	Ug2 outer-bubble charge coupling
k3	1.8	Ug3 string-rotation coupling
k4	2.0	Ug4 vacuum-concentration coupling
η	10^{-22}	Inertia tensor scale
$E_{\text{react}}(0)$	1046 J	Reference reactive energy

A.2 F_U Master Equation (Complete — 4 terms)

$$F_U = U_{g1} + U_{g2} + U_{g3} + U_{g4} + U_{bi} + U_m - \sum_{i=1}^4 |\lambda_i \cdot U_i(r,t)| \cdot E_{\text{react}}$$

Term	Description	Implementation
Ug1	DPM magnetic dipole	<code>compute_Ug1_SOURCE4 / compute_Ug1()</code>
Ug2	Outer-field bubble (charge-reactivity)	<code>compute_Ug2_SOURCE4 / compute_Ug2()</code>
Ug3	Magnetic string rotation	<code>compute_Ug3_SOURCE4 / compute_Ug3()</code>
Ug4	Vacuum concentration (star-BH)	<code>compute_Ug4_SOURCE4 / compute_Ug4()</code>
Ubi	Buoyancy force	<code>compute_Ubi_SOURCE4 / compute_Ubi()</code>
Um	Universal Magnetism (Heaviside-amplified)	<code>compute_Um_SOURCE4 / compute_Um()</code>
$-\sum \lambda_i U_i \cdot E_{\text{react}}$	4th dissipation term (PAPER_420)	<code>compute_FU_SOURCE4 / full pipeline</code>

4th dissipation term parameters (PAPER_420):

$\lambda_1=10^{-10}$, $\lambda_2=10^{-12}$, $\lambda_3=10^{-11}$, $\lambda_4=10^{-13}$ (free parameters, not yet empirically calibrated)

A.3 U_m Heaviside Phase-Transition Amplifier (PAPER_421)

$$U_m^{\text{full}} = U_m^{\text{base}} \times \text{bigl}(1 + 10^{13} \cdot \Theta(\rho_{\text{SCm}} - \rho_c) \text{bigl} \times \text{bigl}(1 + A_q \cos(\Delta \omega a, t) \text{bigl}$$

Symbol	Value	Description
ρ_c	1015 kg/m 3	SCm critical superconducting density

| A_q | 0.1 | Quasi-periodic beating amplitude (10%) |
 | $\Delta\omega$ | $2\pi/(434 \cdot 365.25)$ rad/day | 434-year Gleisberg supercycle |

A.4 UQFF Four Operational Modes

Mode	Dominant Term	Primary Use Case
Compressed	Ug _{sum} + DPM-seeded base	Isolated stellar/BH systems
Resonant	5 resonance frequencies (aDPM, aTHz, ...)	Multi-scale field interactions
Buoyant	$\beta_i U_{bi}$	Expanding nebulae, stellar winds
Superconductive	$U_m (1+10^{13} f_H)$	Magnetars, SCm critical-density regime

Implementation status: all 4 modes operational in MAIN_{1_{CoAnQi}}.cpp*,* CondensedPhysics.py*, and* CondensedPhysics2.py*.*

Appendix: Session 225 Cross-References (PAPER_1000–1081)

> *Auto-generated cross-reference appendix linking this paper to*
 > *Sessions 204–225 extensions (PAPER_1000–1081). Added by*
 > *update_corpus_crossrefs.py (Session 225, April 2026).*

Paper	Title
PAPER_1000	NS Merger F _U Bi Strain Suppression & BCS Gap
PAPER_1001	SMBH Binary Merger F _U Bi Phonon Damping
PAPER_1011	GW170817 NS Merger F _U Bi _i 66.7% Strain Reduction
PAPER_1012	GW190425 Upgraded F _U Bi _i with S26(3)
PAPER_1014	SMBH Merger Inspiral-Coalescence-Ringdown
PAPER_1022	GW Phonon Strain SCm Modulation of h(t)
PAPER_1002	AGN Buoyancy-Corrected Eddington Luminosity
PAPER_1009	3C273 AGN F _U Bi _i Jet Modulation
PAPER_1010	TON618 AGN F _U Bi _i Jet Modulation
PAPER_1037	AGN Buoyancy Jet Calculator — SCm Jet Launching
PAPER_1048	M-Sigma Phonon-Corrected Relation
PAPER_1039	SCm Galaxy Cluster Buoyancy Profile ICM Beta-Model
PAPER_1040	SCm Cluster Merger Shock Mach Number Phonon Damping
PAPER_1041	SCm Cool-Core Buoyancy Balance AGN Feedback
PAPER_1044	SCm Cluster Thermal SZ Effect Compton-y Phonon
PAPER_1045	SCm Cluster Radio Relic Polarization
PAPER_1046	SCm Cluster Lensing Mass Phonon Correction
PAPER_1079	Galaxy Cluster Cooling-Flow Buoyancy Suppression
PAPER_1020	Cosmic Ray Phonon Acceleration DSA Spectrum
PAPER_1035	Kilonova Buoyancy Light Curve r-Process
PAPER_1047	Type Iax Supernova Buoyancy Reversal

21 cross-reference(s) identified.

Appendix: Session 204 Codebase Upgrade Reference

> Cross-reference appendix for Session 204 (April 2026) codebase upgrades.
> Added by upgrade_kozima_ramanujan_appendices.py. For detailed derivations,
> see PAPER_840/851/852/855.

S204.1 Kozima-UQFF LENR Integration

Module	Purpose	Key Result
fneutron_s26_coupling.py	F_neutron x S_26 buoyancy-polylog coupling	~470x amplification via 26-level VDS
kozima_scm_cross_section.py	SCm-modulated neutron-drop cross-section	σ_n^{SCm} with VDS factor $(1 + [\text{SSq}] \cdot n/26)$
kozima_wstp_kernel.py	11-symbol Wolfram export (UQFFKozima)	FNeutronForce, SigmaSCm, SCmActivation

Core equation: $F_{\text{neutron}}^{\text{SCm}} = N_n \sigma_n^{\text{SCm}}(\omega) \Phi_{\text{phonon}}(F_{\{U, Bi\}}/F_U - 1)$
where $\sigma_n^{\text{SCm}}(\omega, n) = \sigma_0 \exp[-(\omega - \omega_{\text{SCm}})^2 / (2\Gamma^2)] (1 + [\text{SSq}] \cdot n/26)$

S204.2 Ramanujan 26-State Summation

Module	Purpose	Key Result
ramanujan_polylog_s26.py	Li_26([SSq]) via Euler-Ramanujan acceleration	15.7+ digits in 53 terms
s26_wstp_kernel.py	8-symbol Wolfram export (UQFFS26)	S26, R26, NaiveLi, S26VDS

Core equation: $S_{26}(z) = \text{Li}_{26}(z) = \eta_{26}(z) / (1 - 2^{\{1-26\}}) + 2^{\{1-26\}} / (1 - 2^{\{1-26\}}) * \text{Li}_{26}(z^2)$

S204.3 Mock Theta Functions (26-State)

Module	Purpose	Key Result
mock_theta_q26.py	f_26(q), phi_26(q), psi_26(q) q-series	Proper q-Pochhammer (a;q)_n

Core equations:

- $f_{26}(q) = \sum_{n=0}^{\{25\}} q^{\{n^2\}} / (-q;q)_{n^2}$
- $\phi_{26}(q) = \sum_{n=0}^{\{25\}} q^{\{n^2\}} / (-q^2;q^2)_n$
- $\psi_{26}(q) = \sum_{n=1}^{\{26\}} q^{\{n^2\}} / (q;q^2)_n$

S204.4 Ramanujan 1/pi with UQFF Modification

Module	Purpose	Key Result
ramanujan_pi_uqff.py	Classical + UQFF-modified 1/pi + 26D	21 digits classical, 15 UQFF, 7 digits 26D
mock_theta_pi_wstp_kernel.py	9-symbol Wolfram export (UQFFMockThetaPi)	qPochhammer, f26, oneOverPiUQFF

Core equation: $1/\pi = (2\sqrt{2}/9801) \sum R_n (1103 + 26390n) W_{26}(n) / C_{26}$
where $W_{26}(n) = \prod_{i=1}^{\{26\}} [1 + [\text{SSq}] \exp(-\kappa_i \cdot n/26)]$

S204.5 Calibration Constants (Canonical)

Symbol	Value	Description
[SSq]	0.57	Universal Quantized Factor
kappa	$5.787 \times 10^{-9} \text{ s}^{-1}$	UQFF exponential decay rate
beta_i	0.603	Buoyancy coupling coefficient
H_SCm	0.99	SCm manifold completeness
rho_SCm	$7.09 \times 10^{-37} \text{ kg/m}^3$	SCm vacuum density
rho_UA	$7.09 \times 10^{-36} \text{ kg/m}^3$	UA aether vacuum density
omega_SCm	$2\pi \times 1.25 \text{ THz}$	SCm phonon resonance
sigma_0	10^{-4}	Base neutron cross-section

Implementation: all modules operational in CondensedPhysics.py, CondensedPhysics2.py, MAIN_{1\CoAnQi}.cpp, and Wolfram kernels (uqff_kozima_kernel.wl, uqff_s26_kernel.wl, uqff_mock_theta_pi_kernel.wl).